



Carboni Bank

Tokenizing Renewable Energy Certification, Hedera Africa Hackathon Oct 25

The Problem

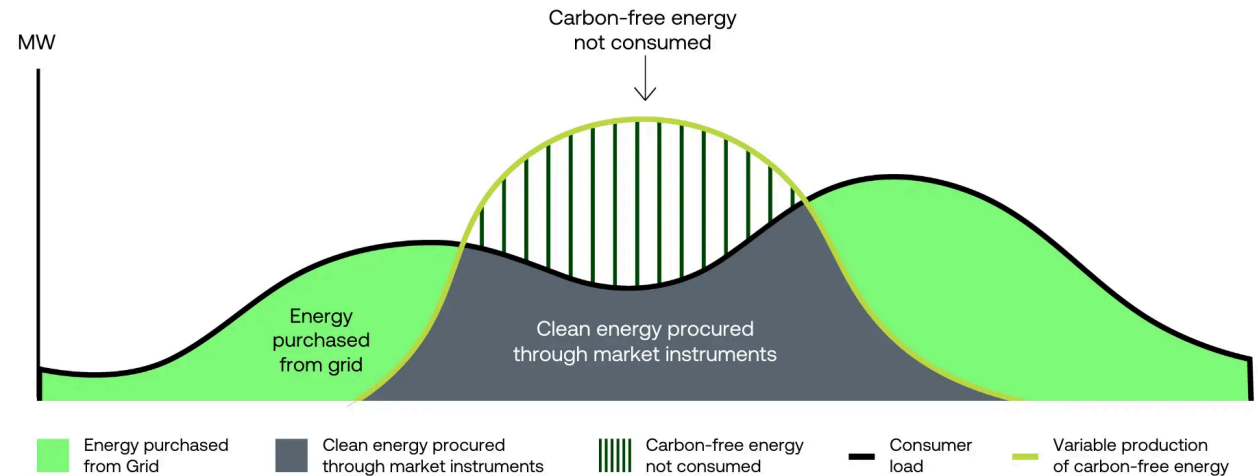
Proving green energy production currently is still mostly a manual process

Current standards are yet to implement sub-hourly granular certification methods, as the current systems use yearly vintages

Consumption verification is not part of the current system, opening a door to false claims

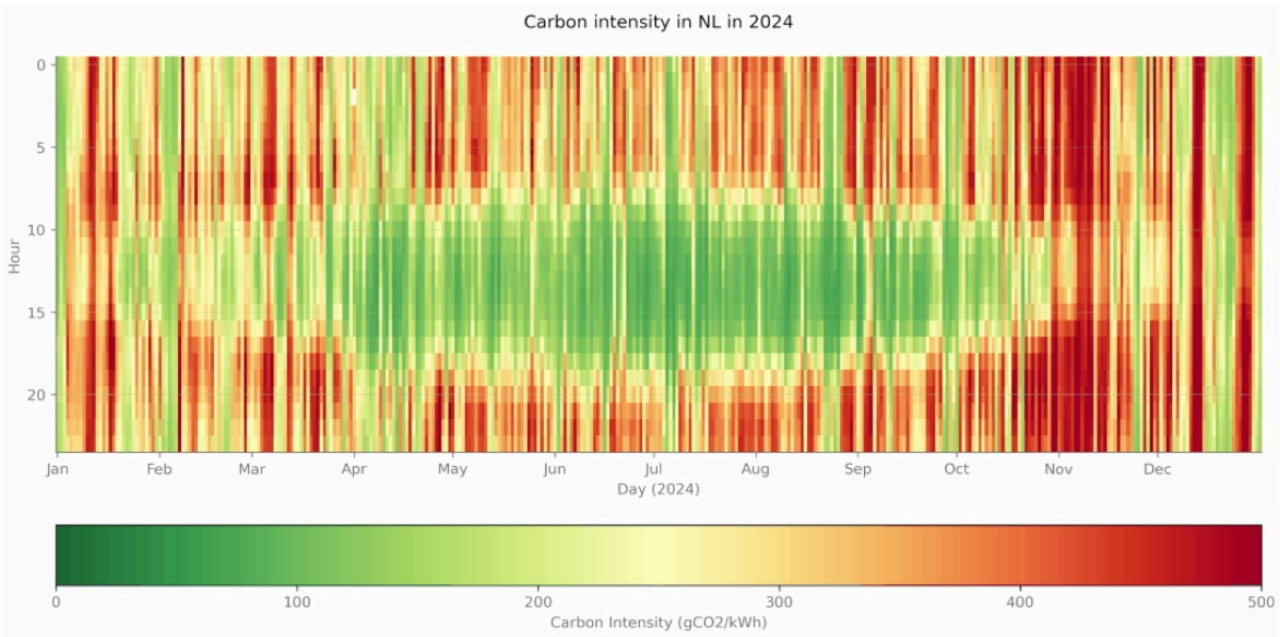
Currently, such energy attributes are only used as proof without any ability to build financial services on top of it

Why hourly matching is important



Adaptation from BCG's report: Granular Scope 2 accounting: Achievable pathways to more accurate emissions reporting

the solution



Depending on supply and demand, the red above represents hours with less available renewables, leading to higher prices, creating a market signal for developing more future renewable supply

By integrating IoT communication to a blockchain based solution we can Automate the verification on both sides of the market at a lower cost including storage.

We can build other financial services on top of it in smart contracts format

The new system allows from the smallest possible granularity starting with hourly and aiming to reach less than 5 min before 2030



Why Hedera

- We wanted a DLT system to build trust
- We wanted a system that we could forecast its operational costs
- We wanted a system that values environmental protection in terms of its carbon footprint
- We want a system that would enable users who never transacted on the web 3.0 to get onboarded easily, in particular managing private keys custody by individuals and entities and the guardian was a really good solution
- The guardian is also focused on sustainability-related matters, which is a great step forward from any currently available services on the market

Market opportunity (Top Down Approach)

according to deep research by chat GPT , which we don't really believe in using such approach

Scenario	TAM	SAM (5-yr)	SOM (5-yr ARR)
Conservative	~\$26.6B	~\$6.0B	~\$6-8M
Aggressive	~\$40-45B	~\$15.6B	~\$140 - 170M

Market opportunity (Buttom up Approch)

our preferred method, we can defend it and map it to the next 18 month operational plan



- **Aggressive 100% market share scenario**

because this a first to market, winner takes all situation

- **~20 TWh/ year**

- Only the Local Egyptian Market**

Carboni has done a proprietary study on the needs of CBAM-related manufacturing for electricity in Egypt alone which can be presented on request

- **20 million Certificates per year needed**

only for CBAM related clients , a certificate is 1 MWh

- **\$20**

The Effective carbon price of each certificate, which is 25% of the current ETS price which is ~80

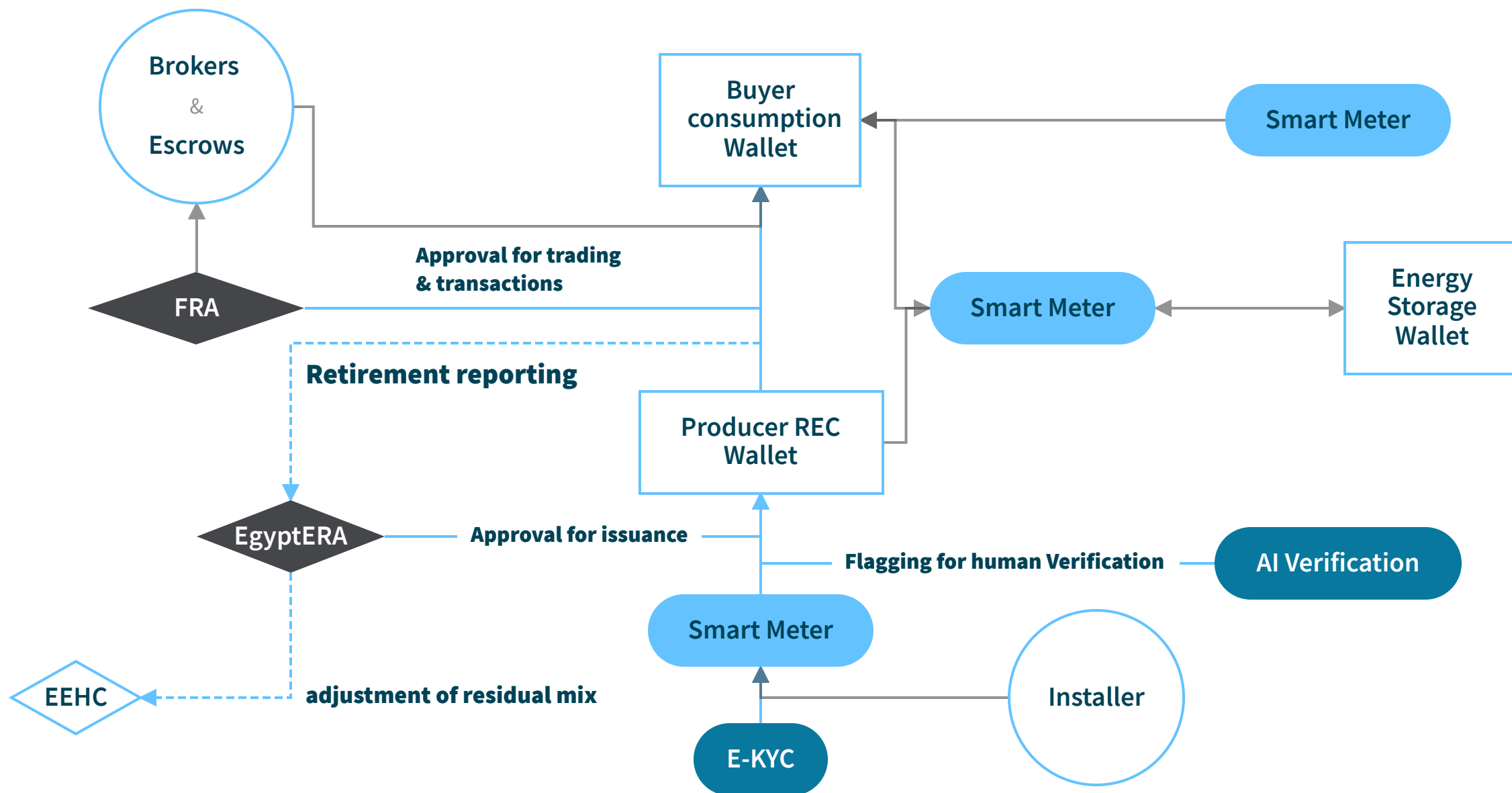
- **\$1**

the lifetime value of each certificate to Carboni

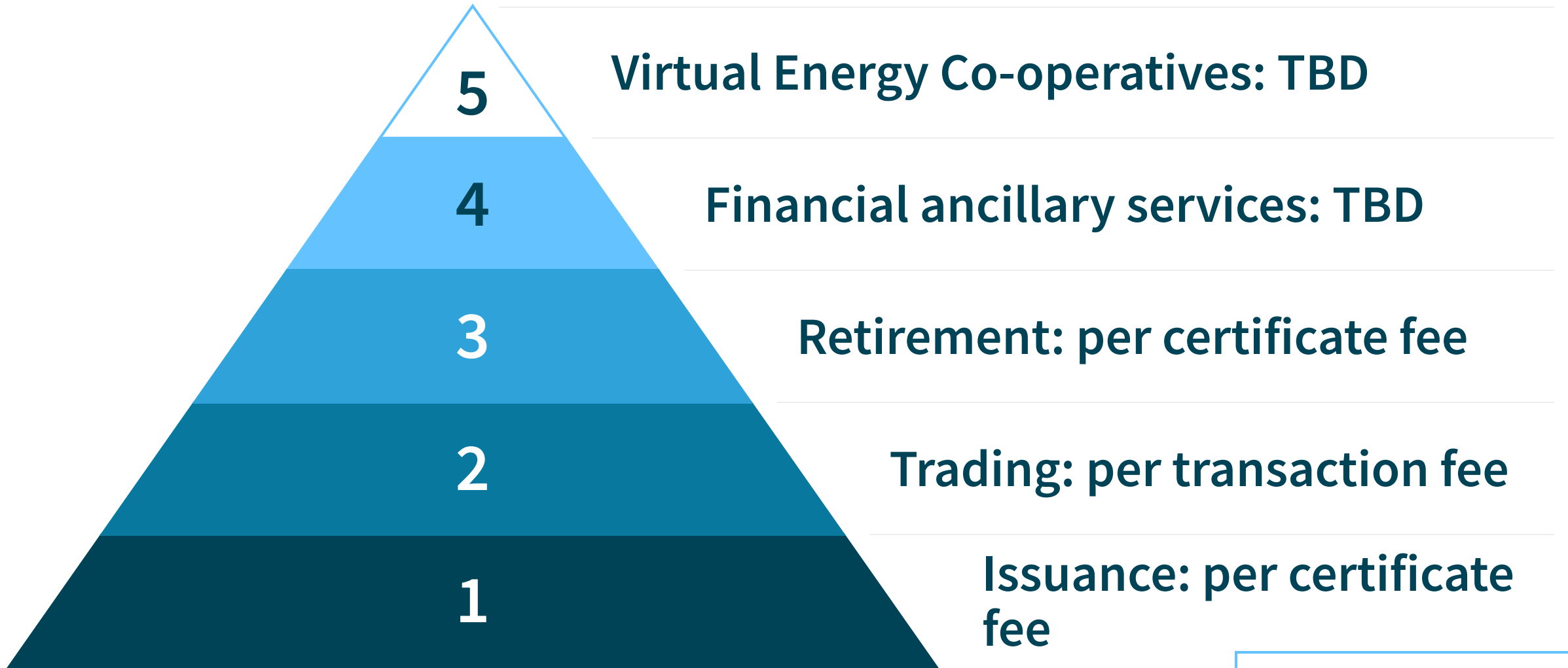
- **\$20 million USD revenues per year**

only form Egypt

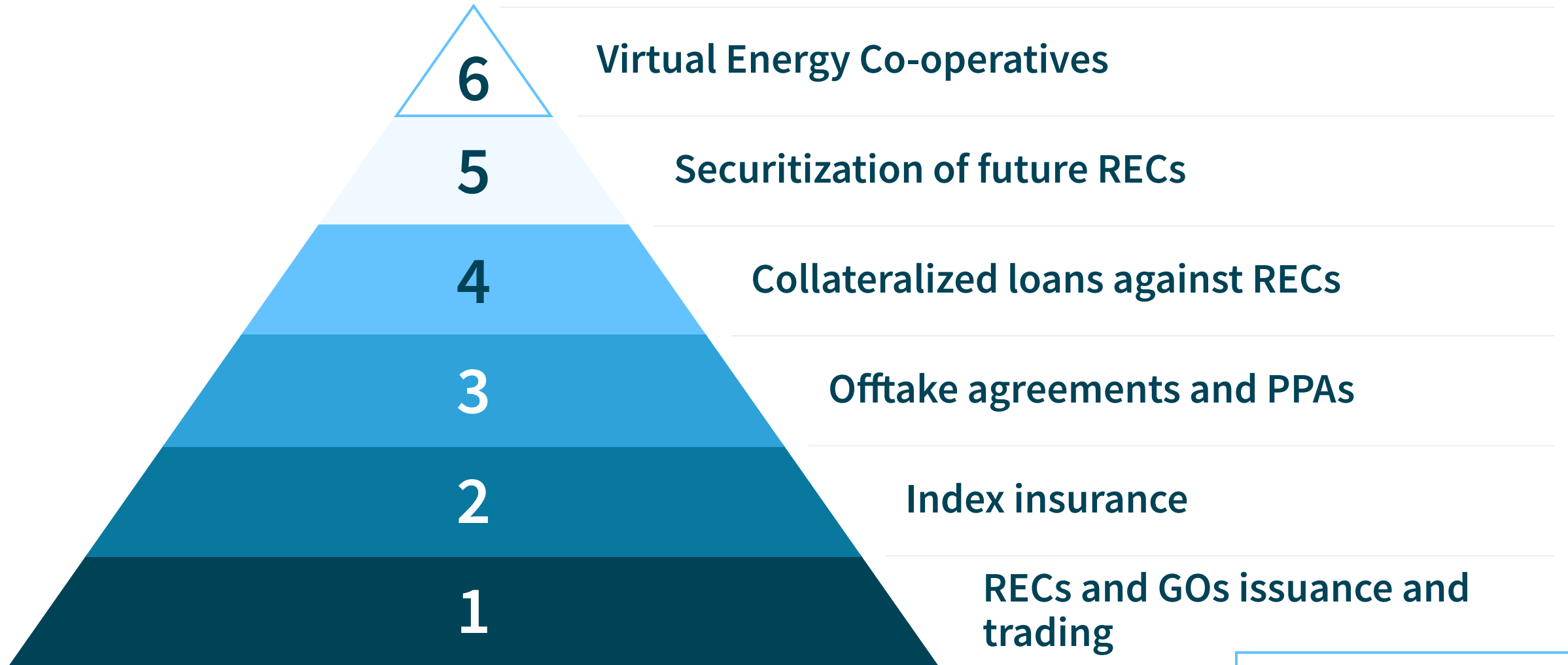
Product Architecture: A Tampering proof Automated Process



Business & Revenue Model



The pyramid of ancillary services on top of RECs



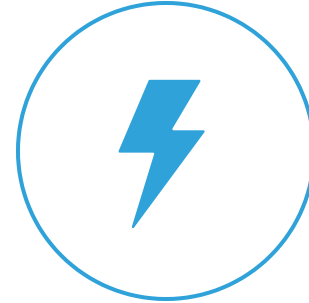


Tokenomics



HBAR

Crypto is banned in Egypt, so we will use the HBAR internally only to pay for gas fees



Renewable energy NFT

Each hour, an NFT is minted and then fractionalized to fungible tokens representing the exact needed energy to allow for the smallest granularity



Stable EGP Coin

Representing the fiat Egyptian pound circulating inside the system

For internal accounting and settling of accounts

The Treasury is managed by Carboni

TRACTION

Comprehensive Startup Progress Report



REGULATORY TRACTION



CLIENT TRACTION



FINANCIAL TRACTION



TECHNOLOGY DEVELOPMENT



Work in Progress

Traction & Milestones 1

Regulatory traction

Carboni has been granted a go signal to develop a PoC by

- **Financial Regulatory Authority (FRA)**
Environmental attribute certificates and Carbon credits were classified as financial instruments last year
- **National Telecommunication Regulatory Authority (NTRA)**
Any company using IoT and accessing the wireless spectrum in Egypt needs approval by the NTRA
- **Egyptian Electric Utility & Consumer Protection Regulatory Agency (EgyptERA)**
regulates the electricity market and responsible for residual mix adjustments,

Financial institutions traction

- **National Bank of Egypt**
is interested in transforming its full renewable energy portfolio to our system and get access to the tokenized attributes
- **Belton**
interested in providing brokerage services to its clients
- **The Egyptian Insurance Federation**
is interested in implementing covers for the producers in our network

TRACTION

Comprehensive Startup Progress Report



REGULATORY TRACTION



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TECHNOLOGY DEVELOPMENT



Work in Progress

Traction & Milestones 2

Clients traction

- **Renewable energy producers**
 - Regional center for renewable energy and energy efficiency (RCREE)
 - manages 1.5 GW of renewable capacity
 - will scale the system to other countries in the MENA region since they are responsible for supporting governments in the region in regards to energy policies
 - NGOs and small producer farmers we have a couple of pilot cases under the 20 KWh
- **Renewable energy offtakers**
 - A telecom company looking to procure 100 GWh (100k certificates) per year
 - A couple of CBAM regulated factories

Technology providers traction

- **Smart meters**

We are starting the PoC with one of the top 5 global smart meters manufacturers and have ongoing discussions with the top BESS producer in the world and other power electronics companies Telecommunication
- **Data SIM Providers**

We are currently in the middle of choosing a provider to provide connectivity to the smart meters

A Team with experience and know how



Dr. Amir Gerges

- Green finance expert and founder of Carboni Bank
- Teaches climate action and carbon markets at the American University in Cairo



Eng. Sameh Abouelsaad

- Software engineer with 5 years of experience specializing in blockchain and backend development.
- passionate about designing scalable systems, writing clean and maintainable code, and delivering solutions that bring real business value

Advisory Board



Dr. Ahmed Darwish (Governance)

- Ex minister in the Egyptian government
- World-renowned expert in the field of the governance of public-private partnership projects
- Ex-chairman of the Suez Canal economic zone



Karim Helal (Investment)

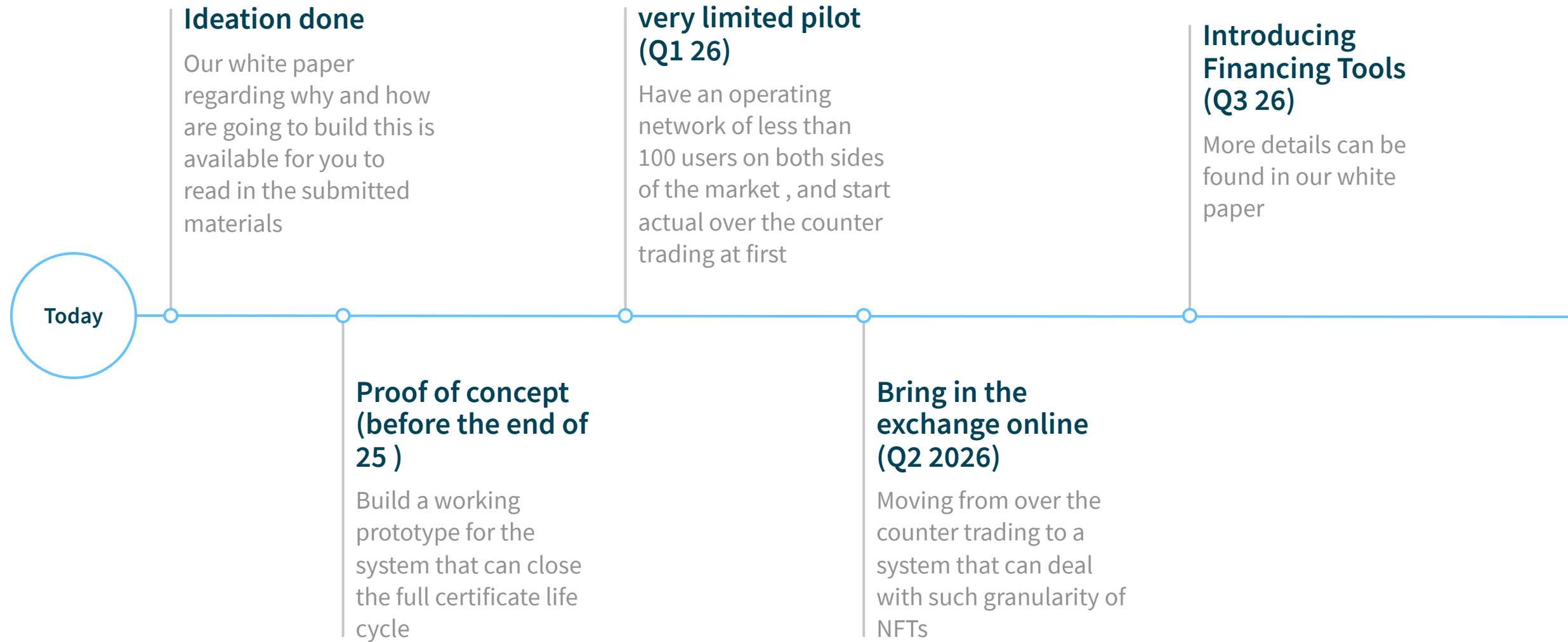
- Ex Chairman of CI Capital, led tens of billion of dollars deals
- Considered one of the God Fathers of investment banking in North Africa



Emad Ghaly (Energy)

- Former CEO of Siemens Egypt who spent over 20 years at Siemens in Germany & Egypt.
- Led several mega projects including the successful 16.4 GW multi-billion Euro Mega Deal in Egypt

Road map





Current Challenges

The Guardian's current architecture

The guardian as is now cant implement the full vision for the system laid out in the white paper , and will to be supplemented with smart contracts and other external services based on the core hedera services like HCS and HTS

Transaction Costs

Forecasting the transaction costs is yet to be done. we know that hedera has a calculator but we are still trying to figure out how using the guardian will affect this

Recent announced increase in gas fees

How the Change from 0.0001 to 0.0008 USD per transaction will affect the transaction costs

Funding

without getting proper funding Carboni will not be able to build this using its own finances



The Ask

Vote for us to win this Hacathon

The cash prize will help us buy time and develop our project further

Technical Support

Given the complexity of adding extra services to the current guardian architecture we will need close mentoring and support by the guardian developers

Funding ~ 1 million USD (18 months runway)

our current estimates to build what we presented is as follows

- monthly salaries 500k (12 people)
- smart meter integrations 100k (starting 2 meter types)
- Operational costs 250k